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**PARIS
SCIENCES
& LETTRES**

Statistical Learning II

Lecture 4 - Least squares

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Summary of ERM

Let $\mathcal{D} = \{(x_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, n\}$ denote training data sampled i.i.d. from p .

Given a choice of:

- Parametric hypothesis class $\mathcal{H} = \{f_\theta : \mathcal{X} \rightarrow \mathcal{Y} : \theta \in \Theta\}$
- Loss function $\ell : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}_+$

Empirical Risk Minimisation consists of:

$$\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_\theta(x_i))$$

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Key questions

- What optimisation procedure to choose?

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- How large n needs to be (with respect to p, d) so that $\hat{\theta} \in \operatorname{argmin} F(\theta)$ has low training and/or test error?
- What properties of the data distribution p makes the problem easier / harder?

Least-squares regression

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Let $\mathcal{D} = \{(x_i, y_i) \in \mathbb{R}^d \times \mathbb{R} : i = 1, \dots, n\}$ denote the training data.

Ordinary least-squares (OLS) regression is defined as:

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \hat{\mathcal{R}}_n(\boldsymbol{\theta}) := \frac{1}{2n} \sum_{i=1}^n (y_i - \langle \boldsymbol{\theta}, \mathbf{x}_i \rangle)^2$$

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$$\min_{\boldsymbol{\theta} \in \mathbb{R}^d} \hat{\mathcal{R}}_n(\boldsymbol{\theta}) := \frac{1}{2n} ||\mathbf{y} - \mathbf{X}\boldsymbol{\theta}||_2^2$$

Where we have defined the data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and label vector $\mathbf{y} \in \mathbb{R}^n$:

$$\mathbf{X} = \begin{bmatrix} - & \mathbf{x}_1 & - \\ - & \mathbf{x}_2 & - \\ & \vdots & \\ - & \mathbf{x}_n & - \end{bmatrix} \in \mathbb{R}^{n \times d} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Bayes risk for OLS

Remarks:

- This corresponds to an ERM problem on the class of linear functions:

$$\mathcal{H} = \{f_{\boldsymbol{\theta}}(\mathbf{x}) = \langle \boldsymbol{\theta}, \mathbf{x} \rangle : \boldsymbol{\theta} \in \mathbb{R}^d\}$$

with the square loss functions:

$$\ell(y, f_{\boldsymbol{\theta}}(\mathbf{x})) = \frac{1}{2} (y - f_{\boldsymbol{\theta}}(\mathbf{x}))^2$$

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- The Bayes predictor and risk are given by:

$$f_{\star}(\mathbf{x}) = \mathbb{E}[y | \mathbf{x}] \quad \mathcal{R}_{\star} = \mathbb{E} \left[\frac{1}{2} (y - \mathbb{E}[y | \mathbf{x}])^2 \right]$$



Exercise:
show this.

Intercept

Remarks:

- Without loss of generality, can add an intercept:

$$f_{\theta}(\mathbf{x}) = \langle \boldsymbol{\theta}, \mathbf{x} \rangle + b$$

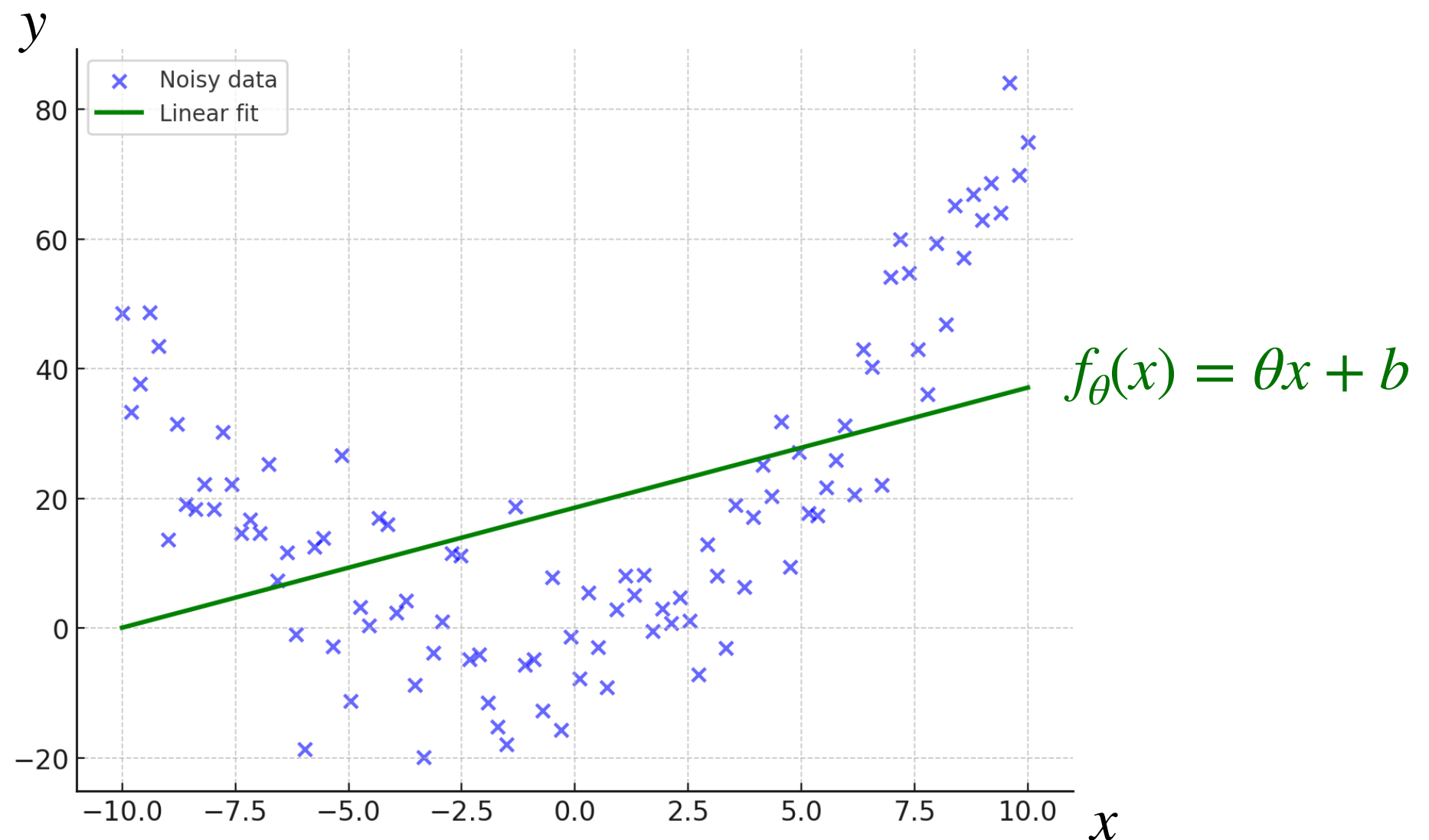
By redefining:

$$\tilde{\mathbf{X}} = \begin{bmatrix} - & \mathbf{x}_1 & - & 1 \\ - & \mathbf{x}_2 & - & 1 \\ & \vdots & & \\ - & \mathbf{x}_n & - & 1 \end{bmatrix} \in \mathbb{R}^{n \times (d+1)}$$

Inductive bias of OLS

Remarks:

- Inductive bias: can only fit affine functions of $\mathbf{x} \in \mathbb{R}^d$



Convexity of OLS

$$\hat{\mathcal{R}}_n(\boldsymbol{\theta}) := \frac{1}{2n} ||\mathbf{y} - \mathbf{X}\boldsymbol{\theta}||_2^2$$

- Gradient: $\nabla_{\boldsymbol{\theta}} \hat{\mathcal{R}}_n = -\frac{1}{n} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) \in \mathbb{R}^d$

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For $n \geq d$, $\hat{\mathcal{R}}_n$ is **strictly convex** if and only if $\text{rank}(\mathbf{X}^\top \mathbf{X}) = d$. This implies that $\hat{\mathcal{R}}_n$ can have at most one global minimum.